

Madeleine Tang Schmidlin (Adams)

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SUMMARY

Experienced process development engineer with expertise in high-temperature materials conversion processes, experimental design, and process risk assessment. Strong record of advancing understanding of material-property-process relationships and applying that knowledge to research and manufacturing challenges. Highly effective in cross-functional environments, partnering with characterization scientists (e.g., XRD, ICP, EPR, XRF, analytical chemistry), composition scientists, plant engineers, technicians, and operators to drive technical insight and practical outcomes. Recognized for technical leadership through invited presentations, mentoring, two filed patents, and 10+ internal technical reports.

EXPERIENCE

Corning Incorporated

Staff Scientist – Melt Process Development, Gorilla Glass (2024 – Present)

Corning, NY

- Led melt process development for advanced glass and glass-ceramic systems for mobile consumer electronics, translating process–structure–property understanding into manufacturable solutions aimed at improved product quality, reduced equipment risk, and informing development strategy.
- Demonstrated that alternative chemical refining agents improved quality in glass-ceramics, eliminating measurable defects while offering potential cost, sustainability, and supply-chain benefits through reduced reliance on 3TG conflict minerals (patent pending). Presented as an invited talk at the ACerS 2026 Spring Meeting.
- Evaluated novel glass and glass-ceramic compositions against critical process contact points to define safe processing windows, aimed at mitigating equipment degradation and protecting core assets from costly refits.
- Applied process-property understanding to establish operating windows reducing inclusion defects by 90% while achieving target material performance and maintaining safe, reliable operation (details protected under Trade Secret).
- Supervised an intern on a melt process development study, defining project scope, providing technical direction, and supporting analysis of process effects on key material properties.

Process Engineer, MT&E – Manufacturing, Technology, & Engineering (2019 – 2024)

Corning, NY

- Provided cross-functional process engineering support across Corning businesses, leading manufacturing scale-up, process development, startup, troubleshooting, and technology evaluation initiatives.
- Assessed multiple glass and glass-ceramic compositions for viability on a sustainable all-electric melting platform. Conducted screening experiments on an induction melter system (patent pending – WO2025174508) and developed a Computational Fluid Dynamics (CFD) model for the same system.
- Accelerated pharmaceutical vial output by launching production of new product for \$20.5M Pfizer order as SKU-change lead. Managed equipment change-over, process development and qualification. Developed plans to minimize downtime.
- Led root cause analyses to determine causes and possible solutions to cosmetic defects through visual inspection, chemical analysis, and experimentation, demonstrating reduced defects on manufacturing line.
- Evaluated \$4M of capital for Corning Pharmaceutical Technologies high-volume expansion through Factory Acceptance Testing (FAT).
- Performed glass forming line debug, documentation, validation, and start-up to produce COVID-19 vaccine vials on an accelerated timeline in support of \$204M government contract (BARDA Operation Warp Speed initiative).

EDUCATION

Cornell University (2022 – 2023)

Ithaca, NY

Master of Engineering in Materials Science and Engineering. Cum Laude.

Oregon State University (2015 – 2019)

Corvallis, OR

Bachelor of Science in Chemical Engineering. Cum Laude.

SKILLS

Process Development | High-Temperature Materials Conversion | Mandarin Chinese (Professional Proficiency) | Statistics | Experimental Design | Six Sigma Green Belt (DMAIC) | Materials & Process Modeling | JMP | Python | FactSage Thermodynamic Modeling | CelSian GTM-X

OTHER ACTIVITIES

- Invited Speaker, American Ceramics Society Glass Manufacturing Symposium (2026)
- Treasurer, Society of Women Engineers – Twin Tiers Section (2020–2022)
- Completed Guthrie Wineglass Half Marathon (2025)